KALASALINGAM ACADEMY OF RESEARCH AND EDUCATION (DEEMED TO BE UNIVERSITY)

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EXTRACTION AND UTILISATION OF LIGNIN AS BIO-ADDITIVE IN A COMPRESSION IGNITION ENGINE FUELLED WITH DIESEL FUEL – AN APPROACH TO IMPROVE FUEL PROPERTIES

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1. Background / Problem Statement

India generates large quantities of agricultural residues, forestry waste andother lignocellulosic biomass, offering potential for sustainableenergy production. Lignin, a carbon-rich aromatic biopolymerinplant cell walls, is investigated in this project as a feedstockforliquid-fuel production through pyrolysis, followed by fuel characterization and evaluation in a compression-ignition (CI) engine.

Research Gap: The project addresses the need to evaluate the physicochemical properties of lignin-derived pyrolysis oil and investigate its suitability for blending with conventional diesel, focusing on engine performance, combustion characteristics and emissions under different operating conditions.

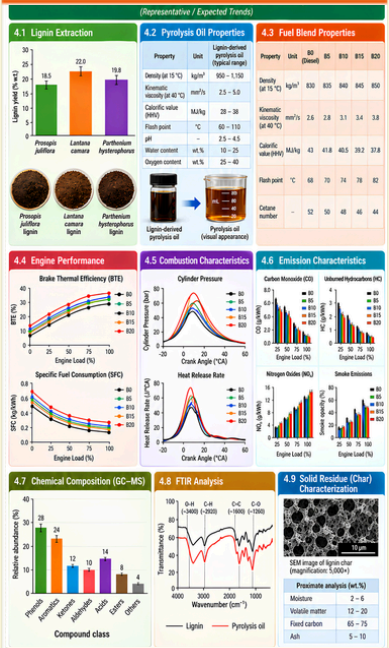
2. Objective(s)

- To extract ligninfrom suitable biomass using an appropriate extraction method.
- To derive bio-additives from lignin through pyrolysis reactions under controlled conditions.
- To evaluate the physical and chemical properties of bio-additive blends with diesel at different dosage levels.
- To investigate the suitability of the prepared blends for use in a compression-ignition (CI) engine.

3. Methodology / Approach



4. Key Results / Findings



5. Innovation / Novelty

-Invasive BiomassValorization: Investigates selected invasive plant biomass as a potential source of lignin for renewable liquid-fuel research.
- Lignin-Focused Conversion: Focuses on extracting lignin and converting it into liquid products through pyrolysis rather than directly using conventional biomass-derived fuel pathways.
- Integrated Fuel Evaluation: Combines pyrolysis-oil characterization, diesel-blend property assessment and compression-ignition (CI) engine testing within a single research framework.
- Controlled Blend Assessment: Evaluates selected blending proportions up to a maximum of 20% lignin-derived liquid fuel to investigate fuel suitability and engine behavior.
- Multi-Parameter Analysis: Examines engine performance, combustion characteristics and emissions together to identify a suitable blend proportion.

Key Innovation: An integrated experimental approach to investigate the potential of lignin extracted from invasive plant biomass for liquid-fuel production and evaluation in diesel blends, connecting biomass utilization with fuel characterization and CI-engine analysis.

6. RE Relevance & Potential Application

- Renewable Energy Development: Investigates the potential of lignin-derived pyrolysis oil as a component of diesel blends, supporting research into alternative liquid fuels.
- Invasive Biomass Utilization: Explores the conversion of selected invasive plant biomass into value-added fuel products, potentially creating additional uses for underutilized biomass resources.
- Reduced Petroleum Dependence: If the resulting blends prove technically suitable, they may offer a pathway to partially substitute conventional diesel.
- Existing Engine Relevance: Evaluates the fuel blends in a compression-ignition (CI) engine to assess their potential compatibility with diesel-engine applications.
- Scalability Potential: The integrated process could be investigated further at pilot scale, subject to lignin recovery efficiency, fuel quality, process energy requirements, economic feasibility and environmental assessment.
- Contribution to Sustainable Energy: Connects invasive biomass utilization, liquid-fuel research and engine performance evaluation within a renewable-energy research framework.

Potential Impact: The study aims to assess whether lignin-derived pyrolysis oil can serve as a technically suitable component of diesel blends, with a maximum blend limit of 20%, while exploring opportunities for biomass valorization and reduced reliance on petroleum-based fuels.

7. Conclusion / Key Takeaways

- Biomass Valorization: The study investigates the potential of lignin extracted from invasive plant biomass as a feedstock for liquid-fuel production.
- Fuel Characterization: Pyrolysis oil will be evaluated for its physicochemical properties to assess its suitability for diesel blending.
- Engine Evaluation: Diesel blends containing up to 20% lignin-derived liquid fuel will be investigated for engine performance, combustion characteristics and emissions.
- Comparative Analysis: The prepared blends will be compared with conventional diesel under different engine operating conditions.
- Sustainable Energy Potential: The research aims to identify a suitable blend proportion while exploring opportunities for biomass utilization and partial substitution of petroleum-based diesel.

Overall Takeaway: This study establishes an experimental framework for evaluating lignin-derived pyrolysis oil as a potential component of diesel blends, with suitability and performance to be determined through testing.

8. References / Acknowledgements

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Acknowledgements

The authors gratefully acknowledge Kalasalingam Academy of Research and Education (KARE) for providing the academic environment and support for this research. The authors also thank their project supervisor and laboratory staff for their guidance and assistance, where applicable.

9. Contact / QR Code

Project Link :

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A0 Portrait (841 × 1189 mm) | PDF | 2–3 columns | Title 60–80 pt | Headings 32–40 pt | Body 24 pt | 300 dpi images preferred

Keep the poster concise, visual and self-explanatory. Prioritise key findings and visuals over large blocks of text.